Advanced Computer Graphics

Light Transport and Path Tracing

Dr. Tom Bashford-Rogers

Monte Carlo Rendering

- Now we have the basics working
  - What about
  - Variance reduction algorithms
  - Realistic cameras
  - Volumes

Monte Carlo Rendering

- Path Tracing: Variance Reduction
  - Many methods
  - Today will discuss two:
    - Better importance sampling
    - Adaptive Sampling

Monte Carlo Rendering

- Path Tracing: Variance Reduction
  - Better importance sampling
    - Can we sample proportional to any terms?
    - Start with cosine term $p(\omega_i) = \frac{\cos(\theta)}{\pi}$
    - We already have this!

$$C(\bar{x}_{i+1}) = C(\bar{x}_{i+1}) \frac{f_r(x_i, \omega_o, \omega_i) \cancel{\cos(\theta)}}{\frac{\cancel{\cos(\theta)}}{\pi}} \qquad C(\bar{x}_{i+1}) = C(\bar{x}_{i+1}) \frac{f_r(x_i, \omega_o, \omega_i)}{\frac{1}{\pi}}$$

Monte Carlo Rendering

- Path Tracing: Variance Reduction
  - Better importance sampling
    - Easy to implement

```
static Vec3 cosineSampleHemisphere(float r1, float r2)

static float cosineHemispherePDF(const Vec3 wi)
```

Monte Carlo Rendering

- Path Tracing: Variance Reduction
  - Adaptive Sampling
  - Some regions of the image contribute more to the variance than others
  - So allocate more samples to those regions
  - Huge literature, but we will focus on a simple approach used in production

Monte Carlo Rendering

- Adaptive Sampling
  - Divide image into blocks
  - Calculate variance within each block $V_{block}(i)$
  - Allocate samples to each block for next iteration

Monte Carlo Rendering

- Adaptive Sampling
  - How to compute variance

$$V_{block}(i) = \frac{1}{N_{block}(i) - 1} \sum_{x,y}^{N_{block}(i)} \left(E[I_{(x,y)}] - I_{(x,y)}\right)^2$$

  - Note $N = W \times H$ and $pixel = (x, y)$
  - But we do not know I_{pixel}

Monte Carlo Rendering

- Adaptive Sampling
  - We have no ground truth
    - Can use mean over region: $I_{(x,y)} \approx \frac{1}{N_{block}(i)} \sum_{x,y}^{N_{block}(i)} E[I_{(x,y)}]$
    - Or can smooth image and use smoothed values:

$$I_{(x,y)} \approx \sum_{i,j} G(i,j) E[I_{(x+i,y+j)}]$$

  - Where $G(i,j)$ is a filter, e.g. a Gaussian

Monte Carlo Rendering

- Adaptive Sampling: Algorithm
  - Render the image at low SPP (1 – 4)
  - Compute variance per block $V_{block}(i)$
  - Calculate weights per block

$$w_i = \frac{V_{block}(i)}{\sum_j V_{block}(j)}$$

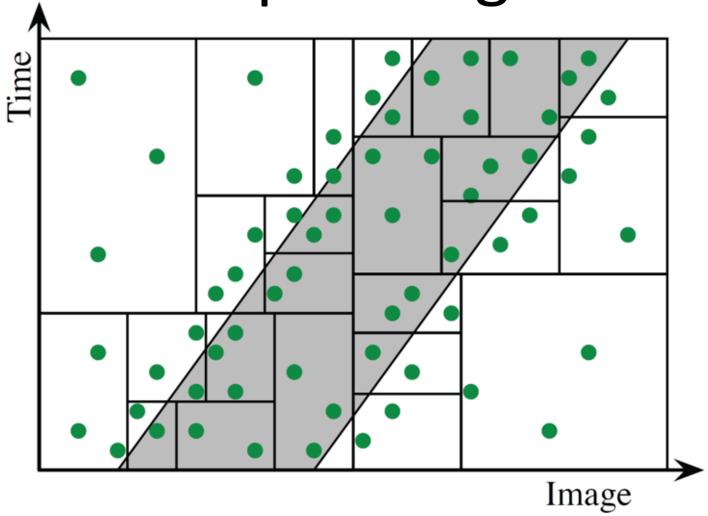
  - Allocate budget of total samples N_{total} to each block $N_i = w_i \cdot N_{total}$

Monte Carlo Rendering

- Adaptive Sampling
  - If all blocks have equal variance
    - Equal number of samples
  - In reality, targets regions with higher variance
  - Must now keep track of SPP per block

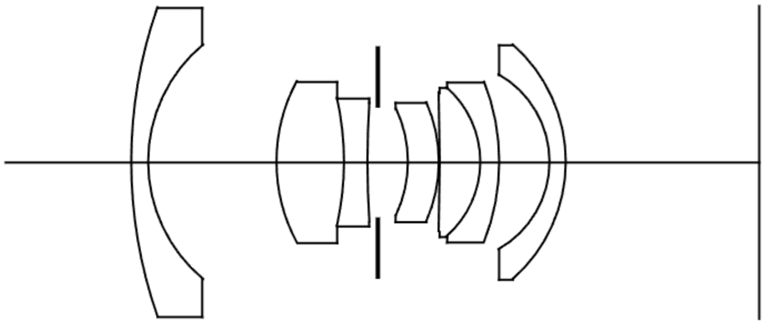
Monte Carlo Rendering

- Adaptive Sampling
  - Can adapt to higher dimensions



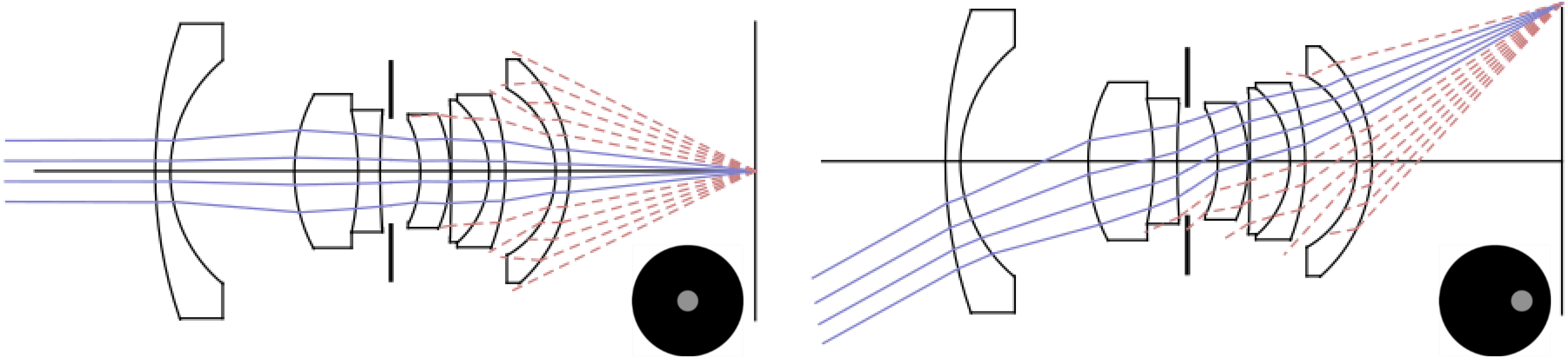
Monte Carlo Rendering

- Realistic Cameras
  - So far used a pin hole camera
  - But ray tracing can represent any camera



Monte Carlo Rendering

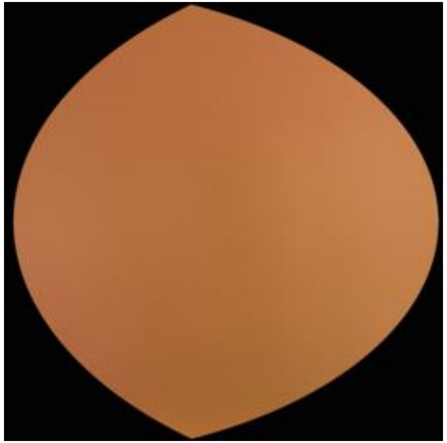
- Realistic Cameras
  - Can sample direction and trace ray through lens system



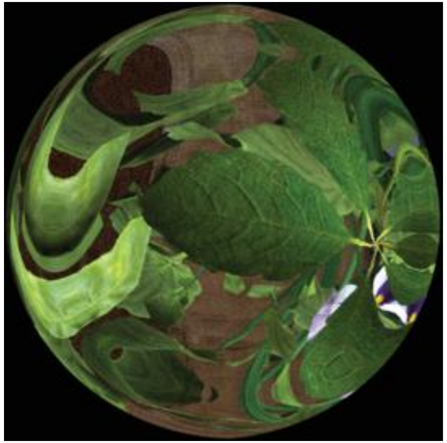
Monte Carlo Rendering

- Realistic Cameras
  - Need to focus the lens, move lens system

Out of Focus



In Focus

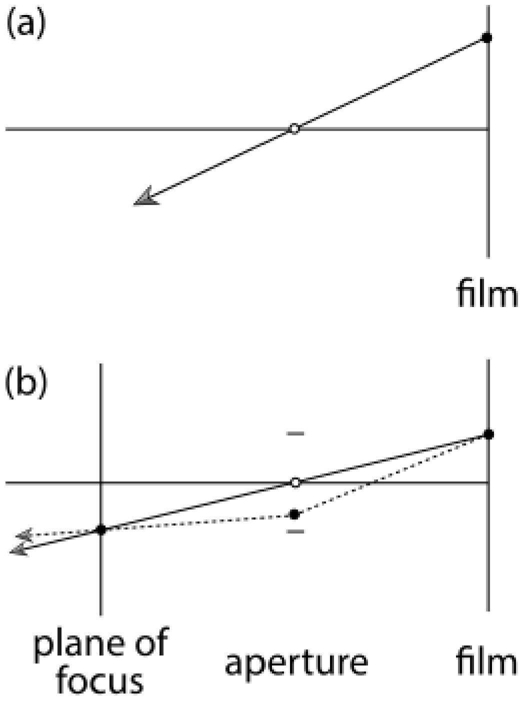


Monte Carlo Rendering

- Realistic Cameras
  - Thin Lens Approximation
  - Simplified lens model
  - Good enough for most applications

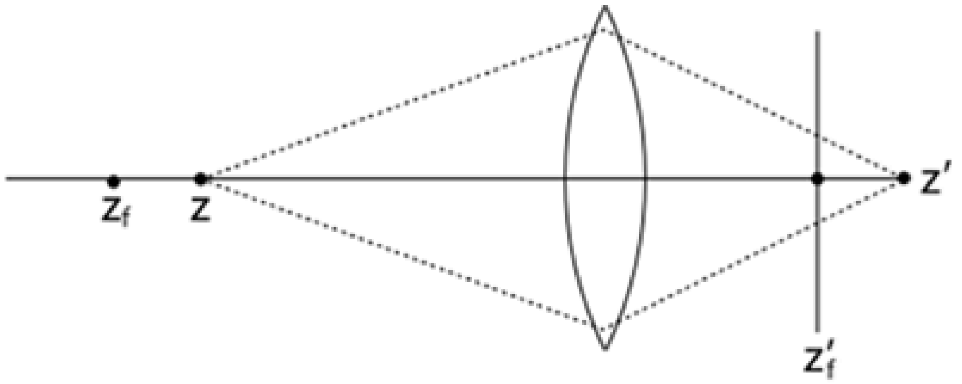
Monte Carlo Rendering

- Realistic Cameras
  - Thin Lens Model
    - Top is pinhole camera
    - Bottom thin lens
      - Infinitely thin lens
      - But still simulates depth of field

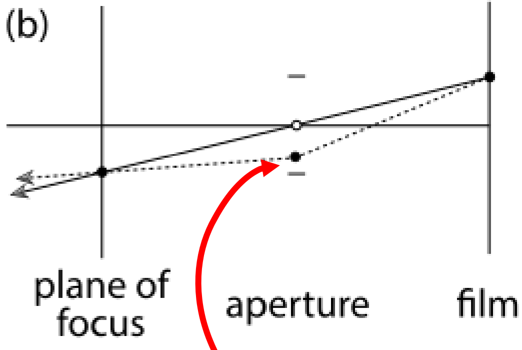


Monte Carlo Rendering

- Realistic Cameras
  - Thin Lens Model
    - Focal distance f_{dist}
    - Lens radius r_{lens}

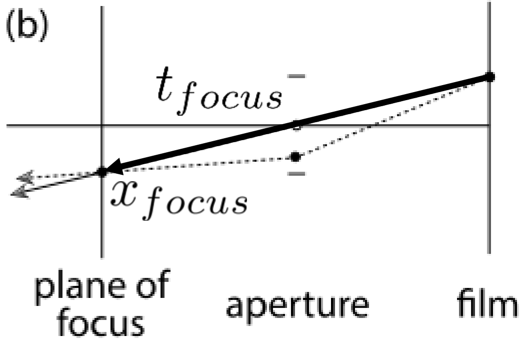


Monte Carlo Rendering



- Realistic Cameras
  - Thin Lens Model
    - Generate ray through pixel $o + t \cdot d$
    - Choose point on lens (sample disk of radius r_{lens}) x_{lens}
    - Intersect ray through centre with focal plane
      - Gives point in focus
    - Update ray origin and direction
    - Start ray from disk in new direction

Monte Carlo Rendering



- Realistic Cameras
  - Thin Lens Model
    - Intersect ray through centre with focal plane

$$t_{focus} = \frac{f_{dist}}{\omega_z}$$

    - Find hitpoint $x_{focus} = o + t_{focus} \cdot d$
    - Update ray $x_{lens} + t \cdot \frac{x_{focus} - x_{lens}}{||x_{focus} - x_{lens}||}$

Monte Carlo Rendering

- Realistic Cameras
  - Extra dimensions to sample
  - More variance
  - Need more samples

Volume Rendering

- So far we have considered surfaces
- What about volumes?

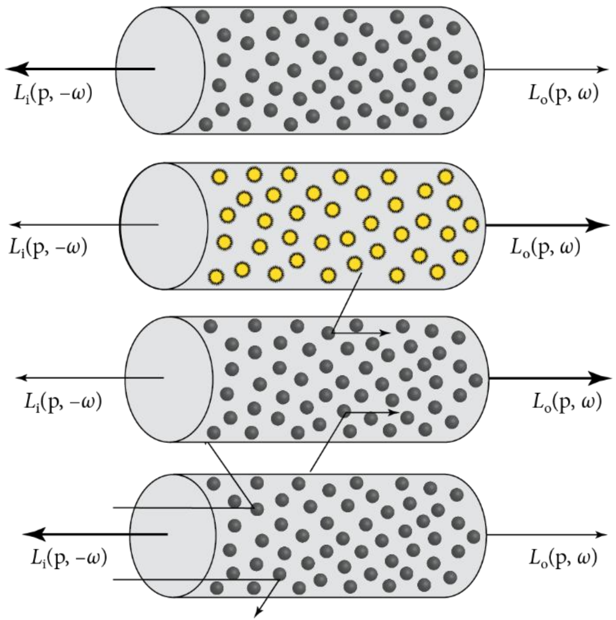


Volume Rendering

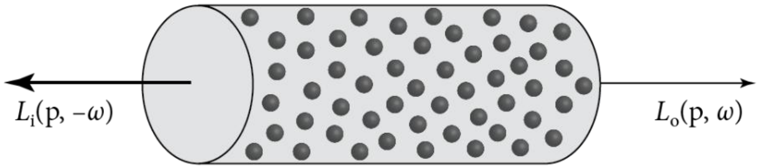
- Participating Media
  - Homogenous vs Heterogenous

Volume Rendering

- Volume Scattering Processes
  – Absorption
  – Emission
  – Inscattering
  – Outscattering



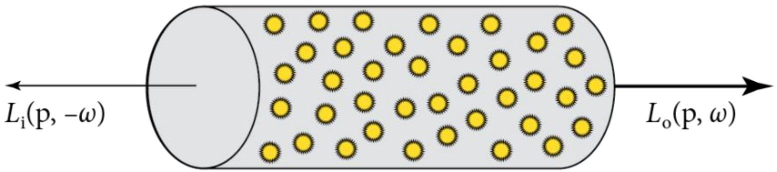
Volume Rendering



- Volume Scattering Processes
  - Absorption
    - Decrease of light due to interactions with particles
    - Specified by absorption coefficient $\sigma_a(x)$
    - Change in radiance for differential length dt :

$$dL_o(x, \omega) = -\sigma_a(x) L_i(x, \omega)$$

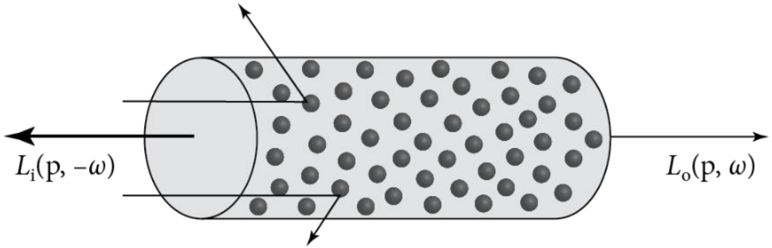
Volume Rendering



- Volume Scattering Processes
  - Emission
    - Increase of light due to emission from particles
    - Specified by $L_e(x, \omega)$

$$dL_o(x, \omega) = \sigma_a(x) L_e(x, \omega)$$

Volume Rendering



- Volume Scattering Processes
  - Out scattering
    - Decrease of light due to scattering from particles
    - Specified by scattering coefficient $\sigma_s(x)$

$$dL_o(x,\omega) = -\sigma_s(x)L_i(x,\omega)$$

Volume Rendering

- Phase Functions
  - Encode how light scatters off particles in a volume: $\rho(\omega_i, \omega_o) = \rho(\cos(\theta))$
    - Where in this case $\cos(\theta) = (\omega_i \cdot \omega_o)$
  - Normalized: $\int_{S^2} \rho(\omega_i, \omega_o) d\omega_i = 1$

Volume Rendering

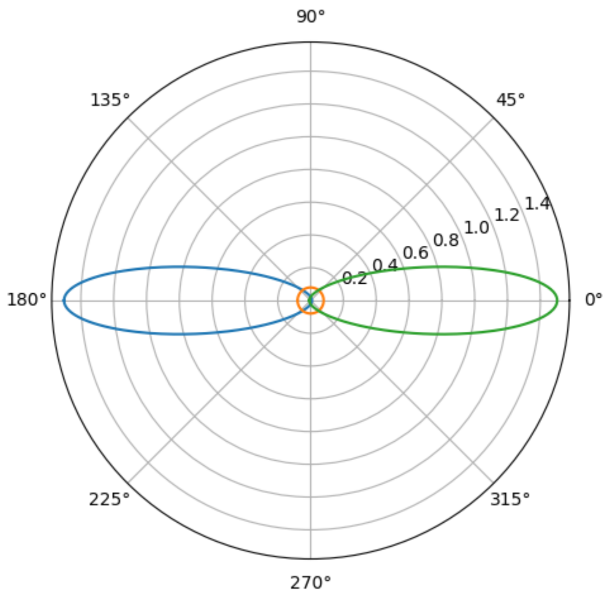
- Phase Functions
  - Isotropic: $\rho(\omega_i, \omega_o) = \dfrac{1}{4\pi}$
  - Henyey-Greenstein
    - Uses an asymmetry parameter g

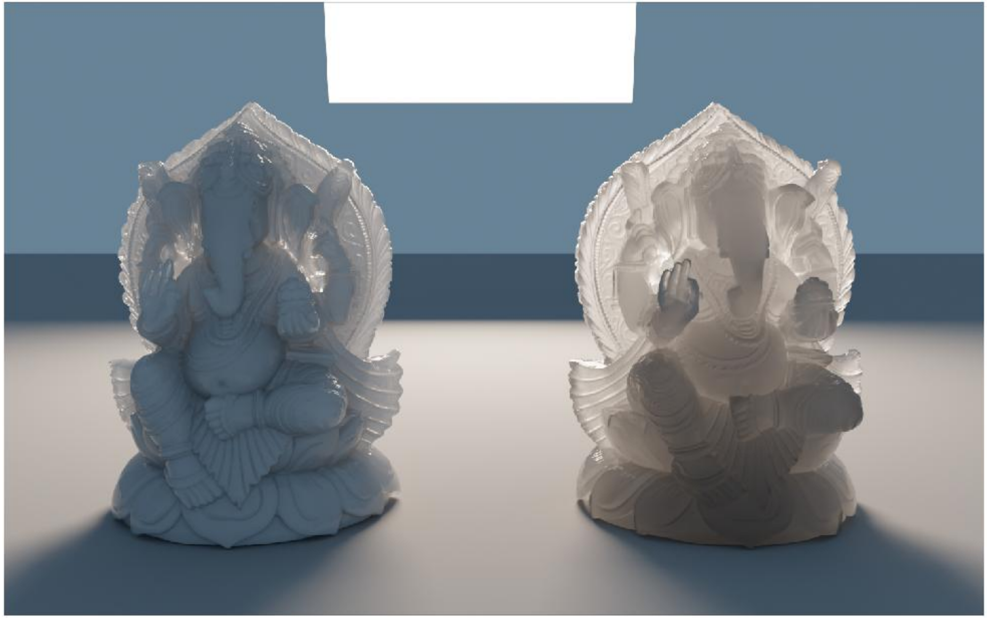
$$\rho(\cos(\theta)) = \frac{1}{4\pi} \frac{1 - g^2}{(1 + g^2 + 2g\cos(\theta))^{\frac{3}{2}}}$$

    - Phase function must be aligned with ω_o : use frame

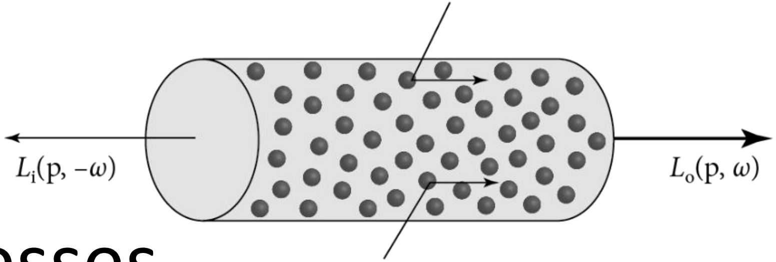
Volume Rendering

– Henyey-Greenstein





Volume Rendering



- Volume Scattering Processes
  - Inscattering
    - Increase of light due to scattering of light from particles into the direction ω

$$\sigma_s(x) \int_{S^2} \rho(x, \omega_o \omega i) L_i(x, \omega_i) d\omega_i$$

Volume Rendering

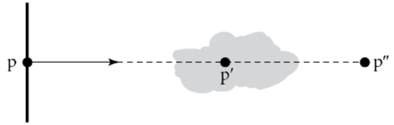
- Combine coefficients
  - Can combine absorption and scattering coefficients
  - Extinction coefficient $\sigma_t(x) = \sigma_a(x) + \sigma_s(x)$
  - Takes into account both absorption and out scattering

Volume Rendering

- Transmittance
  - How much light decreases in the volume

$$\frac{dL_o(x,\omega)}{dt} = -\sigma_t L_i(x,\omega)$$

  - Has solution $T_r(x_i \rightarrow x_{i+1}) = e^{-\int_0^d \sigma_t(x+t\omega)dt}$
  - Known as transmittance
    - Reversable $T_r(x_i \rightarrow x_{i+1}) = T_r(x_{i+1} \rightarrow x_i)$

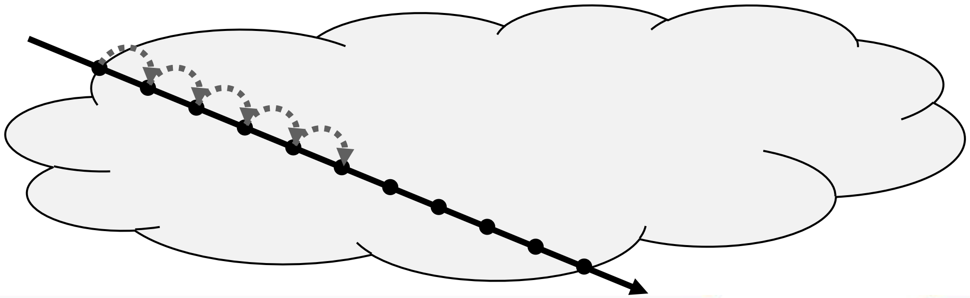


    - Multiplicative $T_r(x_i \rightarrow x_{i+1}) = T_r(x_i \rightarrow x')T_r(x' \rightarrow x_{i+1})$

Volume Rendering

- Transmittance Solutions
  - Homogenous media $T_r(x_i \to x_{i+1}) = e^{-d\sigma_t}$

    Beer's Law
  - Heterogenous Media
    - No closed form solution
    - Ray Marching
      - Biased
    - Estimators



Light Transport

- Rendering Participating Media
  - Incoming light from a direction

$$L_o(x,\omega) = T_r(x_0 \to x)L_o(x_0,-\omega) + \int_0^{t_{max}} T_r(x+t\omega \to x)L_s(x+t\omega,-\omega)dt$$

  - Where $L_s(x,\omega) = \sigma_s(x)\int_{\mathcal{S}^2} \rho(x,\omega_i,\omega_o)L_i(x,\omega_i)d\omega_i$
  - Recursive definition
  - But it is more natural to integrate over paths

Light Transport

- Path Integral Formulation $I = \int_{\mathcal{P}} f(\bar{x}) d\mu(\bar{x})$
  - Consider space of all paths of a fixed length $\mathcal{P}_L$
    - Each space is a cartesian outer product of interaction types

$$\mathcal{P}_L^c = \bigtimes_{i=0}^{L} \begin{cases} \mathcal{M}_i & \text{if } c = 0 \\ \mathcal{V}_i & \text{if } c = 1 \end{cases}$$

$$\mathcal{P}_L = \bigcup_{c \in \{0,1\}^L} \mathcal{P}_L^c$$

Light Transport

- Path Integral Formulation $I = \int_{\mathcal{P}} f(\bar{x}) d\mu(\bar{x})$
  - Consider space of all paths of a fixed length $\mathcal{P}_L$
  - Product Measure now needs to include volume

$$\mu_L^c(D \subseteq \mathcal{P}_L) = \int_D \prod_{i=0}^{L} \begin{cases} dA(x_i) & \text{if } c = 0 \\ dV(x_i) & \text{if } c = 1 \end{cases}$$

  - And $\mu_L(D) = \sum_{c \in \{0,1\}^L} \mu_L^c(D \cap \mathcal{P}_i^c)$
  - Lengths combined as before

Light Transport

- Path Integral Formulation $I = \int_{\mathcal{P}} f(\bar{x}) d\mu(\bar{x})$
  - Phase function can be written as

$$\rho(\omega_i, \omega_o) = \rho(x_{i-1} \rightarrow x_i \rightarrow x_{i+1})$$

  - Path contribution now includes phase functions

$$\hat{f}(x_{i-1} \rightarrow x_i \rightarrow x_{i+1}) = \begin{cases} f_r(x_{i-1} \rightarrow x_i \rightarrow x_{i+1}) & \text{if } x \in \mathcal{M} \\ \sigma_s \rho(x_{i-1} \rightarrow x_i \rightarrow x_{i+1}) & \text{if } x \in \mathcal{V} \end{cases}$$

Light Transport

- Path Integral Formulation $I = \int_{\mathcal{P}} f(\bar{x}) d\mu(\bar{x})$
  - Geometry Term includes volume

$$G(x_i \leftrightarrow x_{i+1}) = \frac{C_s(x_i, x_{i+1}) \cdot C_s(x_{i+1}, x_i)}{||x_{i+1} - x_i||^2} V(x_i \leftrightarrow x_{i+1}) T_r(x_i \leftrightarrow x_{i+1})$$

  - Where

$$C_s(x_i, x_{i+1}) = \begin{cases} \left| \left(\frac{x_{i+1} - x_i}{||x_{i+1} - x_i||} \cdot n_x \right) \right| & \text{if } x \in \mathcal{M} \\ 1 & \text{if } x \in \mathcal{V} \end{cases}$$

  - And $T_r(x_i \leftrightarrow x_{i+1}) = 1$ if in free space

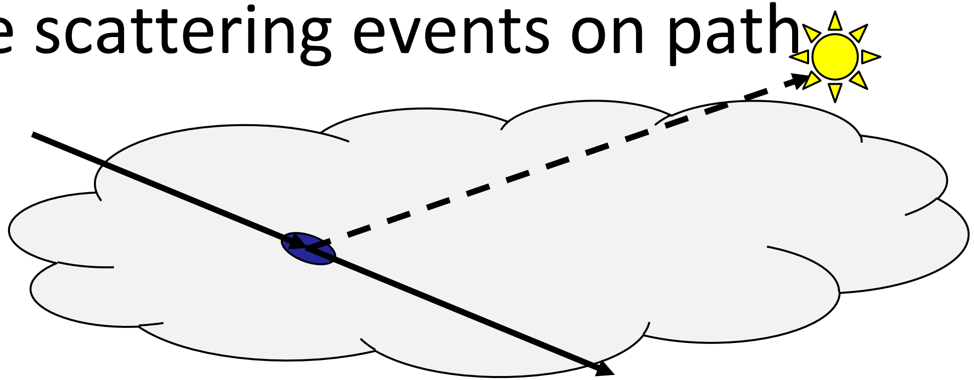
Light Transport

- Path Integral Formulation $I = \int_{\mathcal{P}} f(\bar{x}) d\mu(\bar{x})$
  – Path contribution:

$$f(\bar{x}) = L_e(x_0 \to x_1) \left[\prod_{i=1}^{L-2} G(x_{i-1} \leftrightarrow x_i) \hat{f}(x_{i-1} \to x_i \to x_{i+1}) \right] G(x_{L-1} \leftrightarrow x_L) W_e(x_{L-1} \to x_L)$$

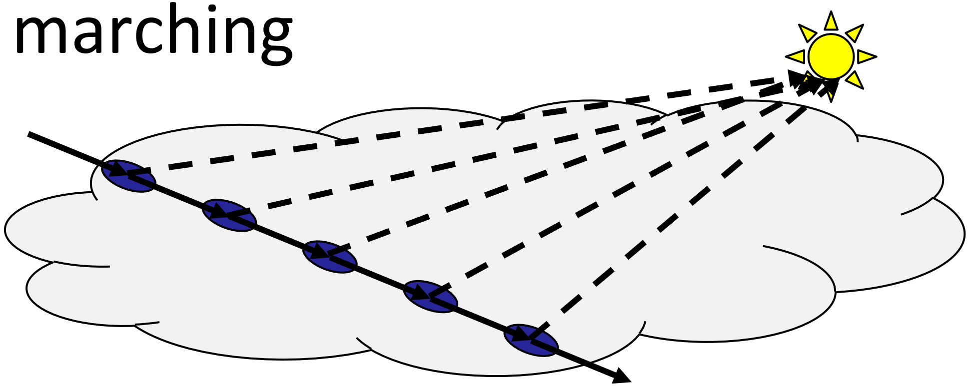
Volume Rendering

- Single Scattering
  - Only compute one scattering event in medium
  - Sample one or more scattering events on path
  - Connect to light



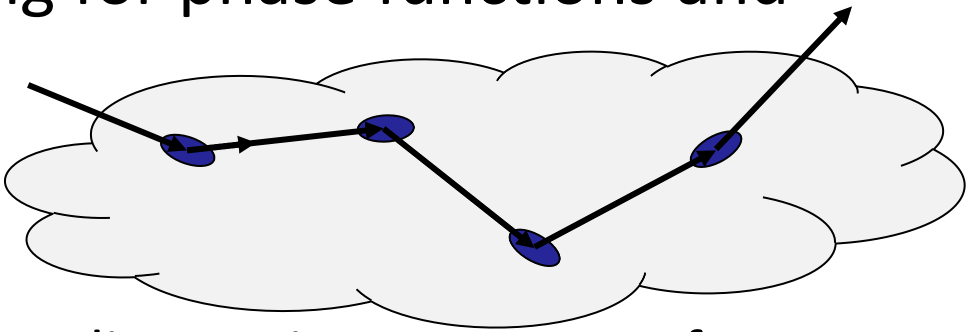
Volume Rendering

- Single Scattering
  - Only compute one scattering event in medium
  - Often used with ray marching



Volume Rendering

- Volumetric Path Tracing
  - Similar to path tracing
  - Except need sampling for phase functions and distance: Sample
    - Distance
    - Phase
    - Repeat until leaves medium or intersects a surface



Volume Rendering

- Single vs Multiple scattering

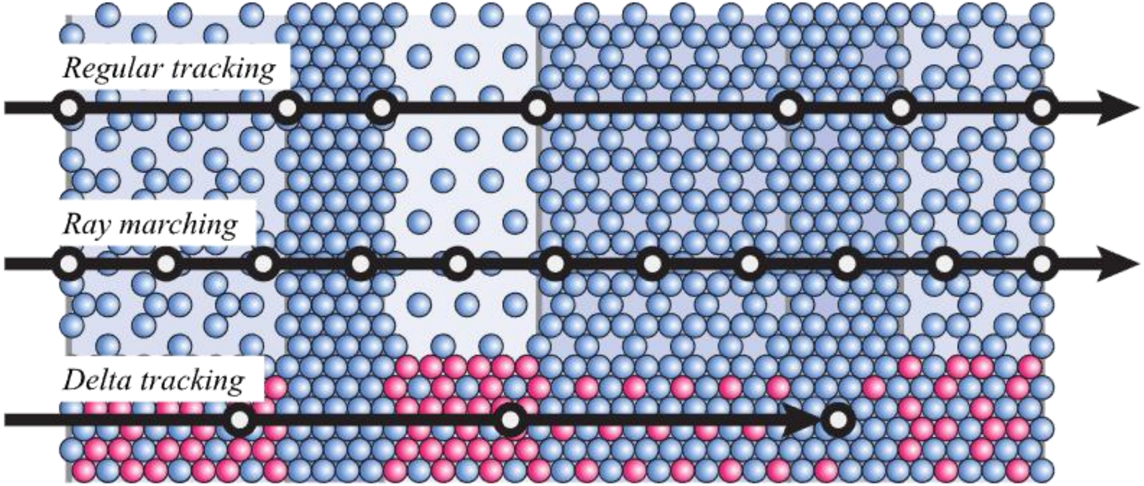
Volume Rendering

- Distance Sampling
  - Homogenous media: $T_r(x_i \rightarrow x_{i+1}) = e^{-d\sigma_t}$
    - Sample distance proportional to transmittance

$$t = \frac{-ln(1-\xi)}{\sigma_t}$$

Volume Rendering

- Heterogenous media
  - Regular Tracking
  - Ray Marching
  - Delta Tracking

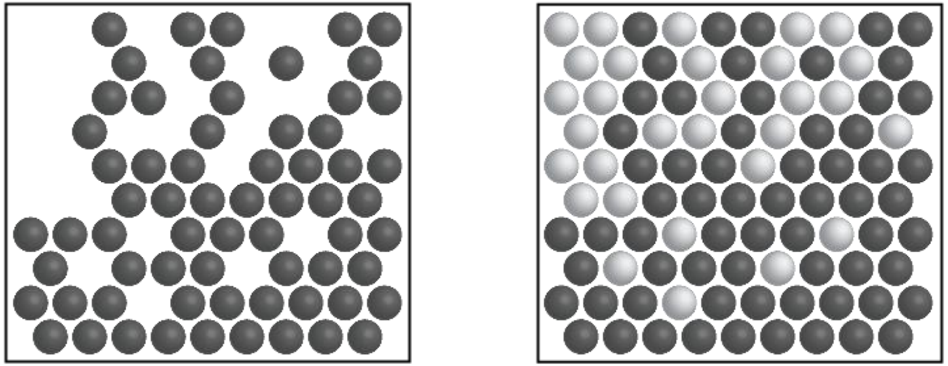


Volume Rendering

- Heterogenous media
  - Delta (Woodcock) Tracking
  - Sample multiple times to pick a distance
  - Prevents sampling location with no medium
  - Unbiased

Volume Rendering

- Heterogenous media: Delta Tracking
  - Needs the majorant $\sigma_{t,maj} = \max(\sigma_t(x)) \quad \forall x \in \mathcal{V}$



Volume Rendering

- Delta Tracking $Delta(o, \omega)$
  - Initialize $t = 0$
  - 1. Sample a distance $t' = -\frac{ln(\xi_s)}{\sigma_{t,maj}}$
  - Add to existing length $t = t + t'$
  - Update position $x = o + t\omega$
  - If $\xi_{s+1} < \frac{\sigma_t(x)}{\sigma_{t,maj}}$ or $x \notin \mathcal{V}$ break
  - Goto 1.

Volume Rendering

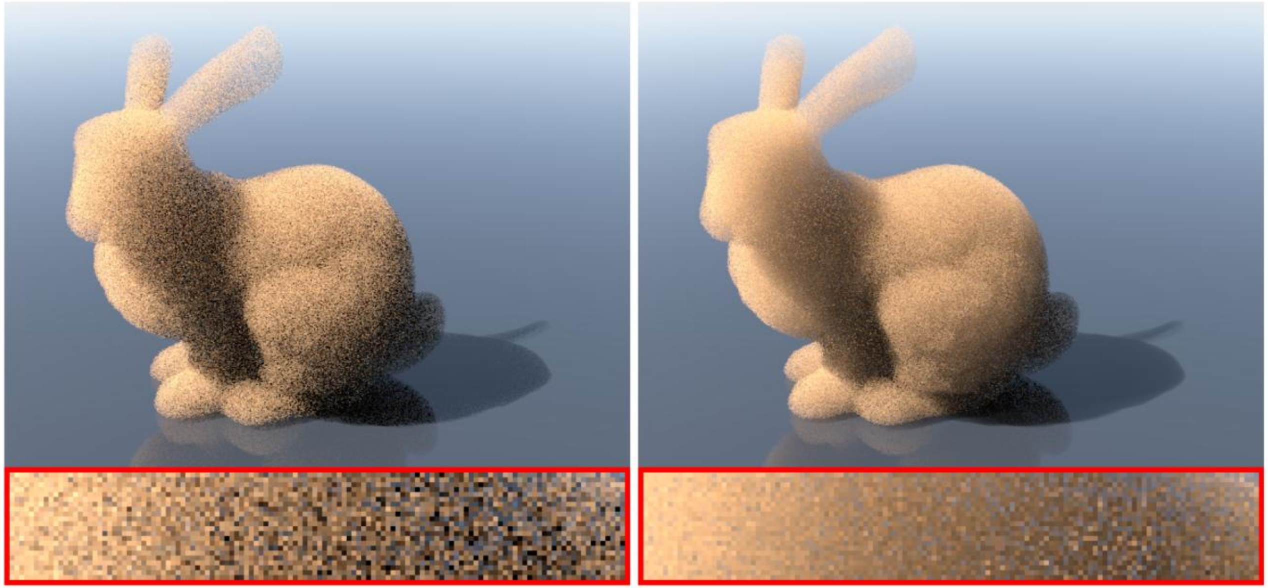
- Heterogenous media
  - Transmittance can be estimated in the same way
    - If $E[T_r(x_i \to x_{i+1})] = \frac{1}{N}\sum_{i=1}^{N}(\text{Delta}(o,\omega) > ||x_i \to x_{i+1}||)$
    - Not very efficient, equivalent to Russian Roulette
  - Can use Ratio Tracking
    - Replace decision to exit with multiplication of continuation probability

Volume Rendering

- Ratio Tracking
  - Initialize $T_r = 1$
  - 1. Sample a distance $t' = -\frac{\ln(\xi_n)}{\sigma_{t,maj}}$
  - Add to existing length $t = t + t'$
  - Update position $x = o + t\omega$
  - Update Transmittance estimate $T_r = T_r \cdot \frac{\sigma_t(x)}{\sigma_{t,maj}}$
  - If $t < t_{max}$ goto 1, otherwise return T_r

Volume Rendering

- Ratio Tracking



Volume Rendering

- Sampling Phase Functions
  - PDF = Phase function as they are normalized
  - Usually sample $\cos(\theta)$ and construct direction
  - Isotropic: $\cos(\theta) = 1 - 2\xi_2$, $\phi = 2\pi\xi_2$
  - Henyey-Greenstein: $\cos\theta = -\frac{1}{2g}\left(1 + g^2 - \left(\frac{1-g^2}{1+g-2g\xi_1}\right)^2\right)$

$$\phi = 2\pi\xi_2$$

Questions?

Computer Graphics

Core Mathematical Concepts

Dr. Tom Bashford-Rogers